AC-90 Rotary Screw Compressor — Operations & Maintenance

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Section 1: Technical Specifications & Operating Parameters

The AC-90 is a single-stage, oil-injected rotary screw air compressor designed for continuous heavy-duty industrial pneumatic service.

Operating Specifications and Limits:

Rated discharge pressure is 8.0 bar (0.80 MPa / 116 psi). The safety pressure relief valve is factory set and certified to open at 10.0 bar (1.0 MPa). Free Air Delivery (FAD) is 9.2 m³/min. Drive motor is 55 kW, 400V 50Hz, 3-phase electric induction motor. Normal operating compressed air discharge temperature is 75°C to 88°C. Maximum trip temperature is 105°C.

Lubrication System: Oil capacity is 25 liters of synthetic polyalphaolefin (PAO) ISO VG 46 rotary screw compressor oil. Oil carryover in delivered air is less than 3 ppm when the separator element is in sound condition.

Section 2: Operating Controls and Alarms

The microprocessor electronic controller continuously monitors air discharge temperature, system pressure, separator differential pressure, and motor running amps.

Alarm Code Matrix:

Alarm H03 — High Discharge Air Temperature: Indicates air/oil mixture discharge temperature exceeded 105°C trip point. Action: Stop the compressor and allow it to cool down completely before inspecting. Check compressor oil level in receiver sight glass. Inspect oil cooler matrix fins for dust or lint accumulation and clean with compressed air reverse blow. Verify thermostatic mixing valve opens at 71°C.

Alarm P02 — High Air Filter Differential: Intake vacuum exceeded 45 mbar. Action: Replace air intake filter cartridge.

Alarm E01 — Main Motor Overload: Drive motor thermal overload trip. Action: Check electrical supply line balance and motor terminal lugs.

Section 3: Maintenance Schedule and Preventative Servicing

Daily Servicing Items:

1. Drain condensate from the receiver tank and moisture separator daily to prevent internal rust and moisture carryover.
2. Check oil level on the receiver sight glass while the compressor is operating fully loaded.
3. Check operating discharge pressure and display panel for advisory messages.

Scheduled Component Replacement Intervals:

Replace Air Intake Filter: Replace the air intake filter cartridge every 2,000 operating hours, or sooner in dusty operating environments.
Replace Oil Filter & Oil: Drain synthetic oil and replace spin-on oil filter element every 4,000 operating hours or every 12 months.
Replace Air/Oil Separator: Replace air/oil separator element every 4,000 operating hours or whenever pressure drop across separator exceeds 1.0 bar.
Drive Belt Inspection: Inspect drive belt tension and alignment every 500 operating hours. Standard belt deflection is 8-10 mm under 50 N center load.